Juan Barrera
Professor Patrick Maher
MBA 544/MIS 545 Applied Data Mining Midterm
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1. Define data mining.

Data mining involves discovering valuable patterns and relationships through statistical methods and machine learning and computational techniques on extensive data collections. The goal of the process is to convert unprocessed information into valuable insights which organizations can use to make decisions.

2. Give one example of an unsupervised learning model.

K-means clustering serves as an example by grouping similar data points into clusters without using any preexisting labels.

3. Give one example of a supervised learning model.

Logistic regression demonstrates its ability to predict categorical results by using labeled data to determine whether something is classified as spam or not spam.

4. Explain the difference between classification and regression models.

The two model types use different prediction methods because classification models show which category to choose while regression models estimate exact numbers.

5. Explain Market Basket Analysis.

Market Basket Analysis uses association rules to find connections between products which customers tend to purchase together. Customers who purchase bread usually buy butter.

6. Define clustering and compare with classification.

Clustering serves as an unsupervised technique which creates data groups through similarity assessment without requiring existing category designations. Classification functions as a supervised system which utilizes labeled training data to distribute data into specific predefined categories.

7. What does Pearson correlation measure?

Pearson correlation assesses the strength and direction of a linear relationship which exists between two continuous variables through a range that extends from -1 to +1.

8. What is the difference between supervised and unsupervised learning?

Supervised learning requires labeled data for model training, whereas unsupervised learning discovers patterns within data that lacks labels.

9. What is model induction?

Model induction involves the task of creating a general model which describes patterns by studying specific training data through the process of discovering rules and relationships.

10. Describe kNN and why it is a lazy learner.

The k-Nearest Neighbors algorithm classifies data points by using the most common label from their closest neighboring data points. The system operates as a lazy learner because it only computes predictions when required after storing data without creating a model during the training phase.

11. Define overfitting.

Overfitting describes a situation where a model becomes unable to perform well on new data because it memorizes training data including its noise components.

12. What is a term-document matrix?

The term-document matrix functions as a text data representation which displays terms (words) in rows and documents in columns while showing term frequency or importance through its value system.

13. Explain nodes, edges, and attributes in networks.

- **Nodes:** exist as entities which include people within a social network.
- **Edges:** exist as relationships which connect different nodes through friendships.
- **Attributes:** provide extra details about nodes and edges which include attributes like age and weight.

14. Which one is an unsupervised learning technique?

A) Market Basket Analysis

15. Which one is a supervised learning technique?

B) Logistic Regression

16. Which one is a classification technique?

C) Logistic Regression

17. Congrats, you got this far!

Thank you Professor!